

VoiceOps - Multi-Domain AI Automation Agent

Project Vision

VoiceOps is a voice-enabled autonomous AI agent designed to automate personal operations across finance tracking, file organization, learning management, and life administration. The system will demonstrate tool calling, MCP server integration, GitHub Actions automation, structured outputs, and cloud deployment readiness.

Core Domains & Features

- 1. Expense Tracker Automation: Add expenses via voice, generate summaries, detect budget anomalies, produce monthly PDF reports.
- 2. Intelligent File Organizer: Scan directories, categorize files using LLM classification, auto-move files, summarize PDFs.
- 3. AI Learning Assistant: Generate weekly AI learning plans, summarize research papers, maintain progress tracker.
- 4. Smart Life Admin Assistant: Create reminders, schedule calendar events, send automated weekly summaries via email.

Technical Architecture & Tools

- Voice Layer: Whisper (Speech-to-Text) + TTS (pyttsx3 or ElevenLabs).
- Backend: FastAPI-based agent orchestrator.
- LLM: GPT model with structured tool calling.
- Tool Registry: MCP server exposing local tools safely.
- Storage: CSV (initial), scalable to PostgreSQL or DynamoDB.
- Automation: GitHub Actions for report generation & CI/CD workflows.
- Optional Cloud Deployment: AWS Lambda + S3 + SES + Transcribe + Polly.

Planned Tool Interfaces (Function Calling Schema)

- `add_expense(category, amount, date)`
- `get_monthly_summary(month)`
- `organize_directory(path)`
- `summarize_pdf(file_path)`
- `create_calendar_event(title, datetime)`
- `send_email(recipient, subject, body)`
- `generate_learning_plan(topic)`

Implementation Roadmap

- Phase 1: CLI agent with expense tool calling and structured JSON outputs.
- Phase 2: Add file organizer and learning assistant tools.
- Phase 3: Integrate voice input/output layer.
- Phase 4: Implement MCP server and GitHub Actions automation.
- Phase 5: Deploy cloud-ready version on AWS.

Outcome: A production-grade multi-domain AI automation system suitable for portfolio demonstration, research experimentation, and real-world personal productivity enhancement.